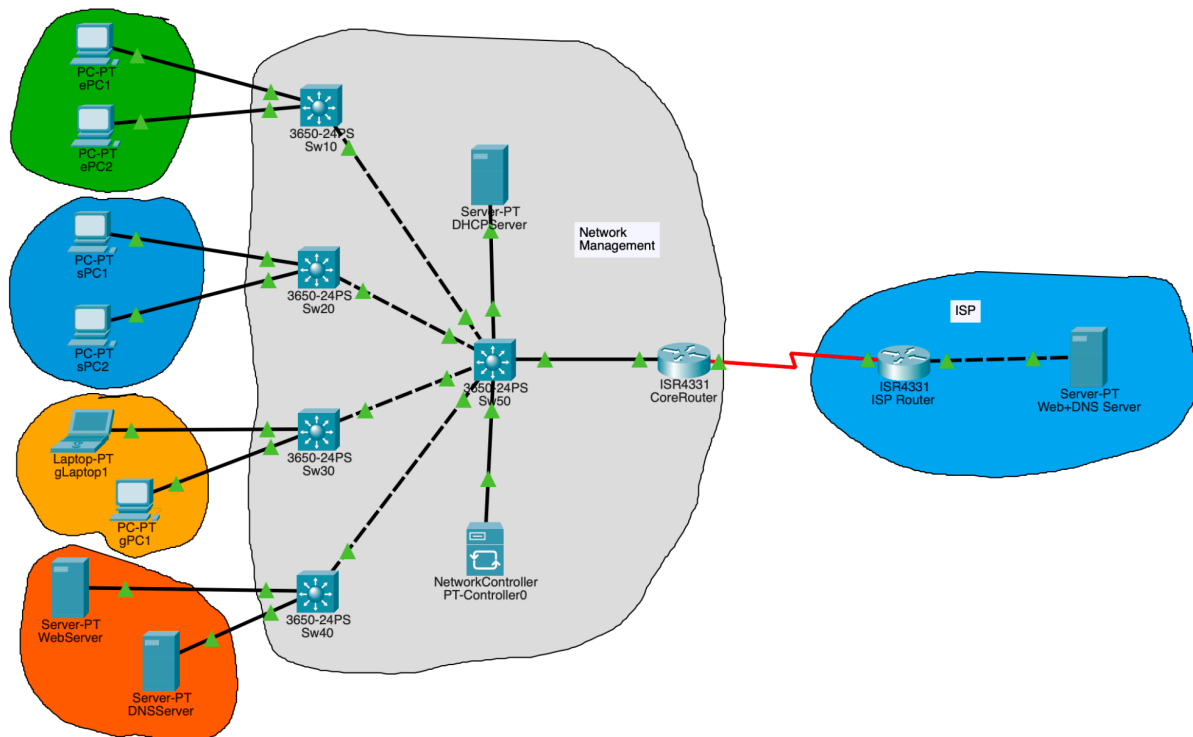


Secure Network Design Configuration

This project is designed to demonstrate the configuration of an enterprise network. The network utilizes network segmentation via virtual local area network (VLAN) to isolate traffic, and restricts broadcast domains. A Dynamic Host Configuration Protocol (DHCP) server is set up to automatically allocate IP addresses, gateways, subnet masks, and lease time. A Domain Name System (DNS) server to resolve domain names to IPs and allow for previous DNS queries to be cached, allowing for users to experience faster connection to websites. A Web server to host the website of the enterprise. And the Web server and DNS server are placed in a demilitarized zone (DMZ) with restricted access because they are forward facing to the internet. Switches that have Access Control Lists (ACL) bound to them to control network traffic. Finally, a network controller to monitor subnets, apply ACLs, and optimize network traffic.

User Groups	VLAN	IP address Range	Netmask
Executive	10	10.12.10.0/24	255.255.255.0
Employee	20	10.12.20.0/24	255.255.255.0
Guest	30	10.12.30.0/24	255.255.255.0
DMZ	40	10.12.40.0/24	255.255.255.0
Network Management	50	10.12.50.0/24	255.255.255.0

Topology of the Network



1. Task1 - Connectivity

a. VLAN configuration of Sw10-50

```
Sw10#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Gig1/0/3, Gig1/0/4, Gig1/0/5, Gig1/0/6, Gig1/0/7, Gig1/0/8, Gig1/0/9, Gig1/0/10, Gig1/0/11, Gig1/0/12, Gig1/0/13, Gig1/0/14, Gig1/0/15, Gig1/0/16, Gig1/0/17, Gig1/0/18, Gig1/0/19, Gig1/0/20, Gig1/0/21, Gig1/0/22, Gig1/0/23, Gig1/1/1, Gig1/1/2, Gig1/1/3, Gig1/1/4
10	Executive	active	Gig1/0/1, Gig1/0/2
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

i.

Sw20#show vlan brief

VLAN	Name	Status	Ports
1	default	active	Gig1/0/3, Gig1/0/4, Gig1/0/5, Gig1/0/6 Gig1/0/7, Gig1/0/8, Gig1/0/9, Gig1/0/10 Gig1/0/11, Gig1/0/12, Gig1/0/13, Gig1/0/14 Gig1/0/15, Gig1/0/16, Gig1/0/17, Gig1/0/18 Gig1/0/19, Gig1/0/20, Gig1/0/21, Gig1/0/22 Gig1/0/23, Gig1/1/1, Gig1/1/2, Gig1/1/3 Gig1/1/4
20	Employee	active	Gig1/0/1, Gig1/0/2
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

ii.

Sw30#show vlan brief

VLAN	Name	Status	Ports
1	default	active	Gig1/0/3, Gig1/0/4, Gig1/0/5, Gig1/0/6 Gig1/0/7, Gig1/0/8, Gig1/0/9, Gig1/0/10 Gig1/0/11, Gig1/0/12, Gig1/0/13, Gig1/0/14 Gig1/0/15, Gig1/0/16, Gig1/0/17, Gig1/0/18 Gig1/0/19, Gig1/0/20, Gig1/0/21, Gig1/0/22 Gig1/0/23, Gig1/1/1, Gig1/1/2, Gig1/1/3 Gig1/1/4
30	Guest	active	Gig1/0/1, Gig1/0/2
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

iii.

show vlan brief

VLAN	Name	Status	Ports
1	default	active	Gig1/0/3, Gig1/0/4, Gig1/0/5, Gig1/0/6 Gig1/0/7, Gig1/0/8, Gig1/0/9, Gig1/0/10 Gig1/0/11, Gig1/0/12, Gig1/0/13, Gig1/0/14 Gig1/0/15, Gig1/0/16, Gig1/0/17, Gig1/0/18 Gig1/0/19, Gig1/0/20, Gig1/0/21, Gig1/0/22 Gig1/0/23, Gig1/1/1, Gig1/1/2, Gig1/1/3 Gig1/1/4
40	DMZ	active	Gig1/0/1, Gig1/0/2
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

iv.

Sw40#

Sw50#show vlan brief

VLAN	Name	Status	Ports
1	default	active	Gig1/0/6, Gig1/0/7, Gig1/0/8, Gig1/0/9 Gig1/0/10, Gig1/0/11, Gig1/0/12, Gig1/0/13 Gig1/0/14, Gig1/0/15, Gig1/0/16, Gig1/0/17 Gig1/0/18, Gig1/0/19, Gig1/0/20, Gig1/0/21 Gig1/0/22, Gig1/1/1, Gig1/1/2, Gig1/1/3 Gig1/1/4
10	Executive	active	
20	Employee	active	
30	Guest	active	
40	DMZ	active	
50	NetMgmt	active	Gig1/0/5, Gig1/0/23
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

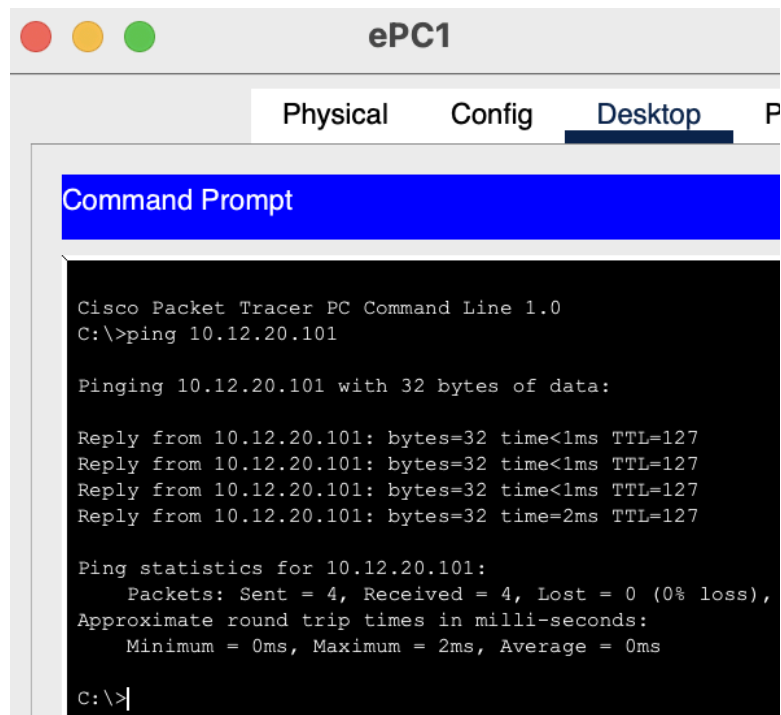
- b. List of the Gateways IPs for all the VLANs from core router

```
CoreRouter#show ip interface brief
```

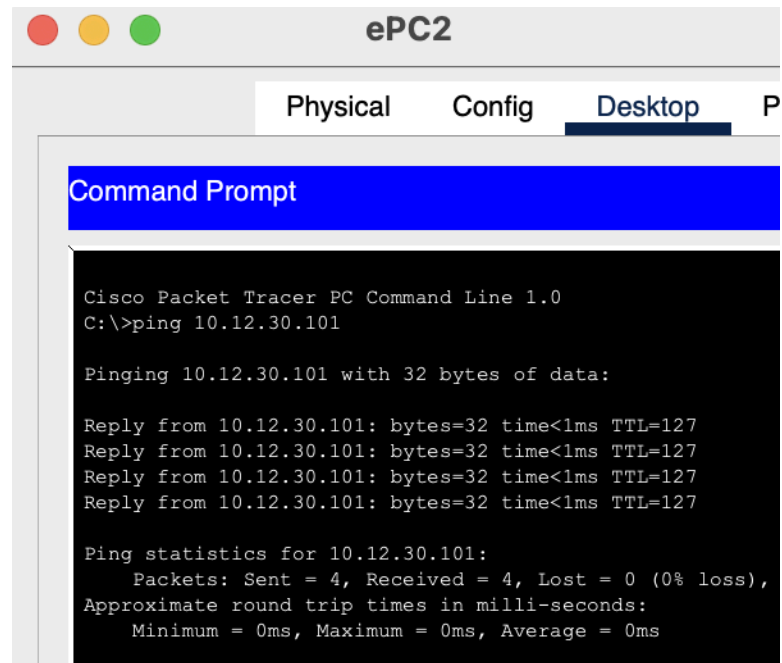
Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0/0	unassigned	YES	NVRAM	administratively down	down
GigabitEthernet0/0/1	unassigned	YES	NVRAM	administratively down	down
GigabitEthernet0/0/2	unassigned	YES	NVRAM	administratively down	down
GigabitEthernet0/1/0	unassigned	YES	unset	up	up
GigabitEthernet0/1/1	unassigned	YES	unset	down	down
GigabitEthernet0/1/2	unassigned	YES	unset	down	down
GigabitEthernet0/1/3	unassigned	YES	unset	down	down
Serial0/2/0	200.12.1.2	YES	manual	up	up
Serial0/2/1	unassigned	YES	unset	down	down
Vlan1	unassigned	YES	unset	administratively down	down
Vlan10	10.12.10.1	YES	manual	up	up
Vlan20	10.12.20.1	YES	manual	up	up
Vlan30	10.12.30.1	YES	manual	up	up
Vlan40	10.12.40.1	YES	manual	up	up
Vlan50	10.12.50.1	YES	manual	up	up

- c. Ping requests (not yet demonstrating the access control list, just the ability of the devices to ping each other)

- i. Exec to Employee



ii. Exec to Guest



The screenshot shows a Cisco Packet Tracer PC window titled "ePC2". The "Desktop" tab is selected. A "Command Prompt" window is open, displaying the following text:

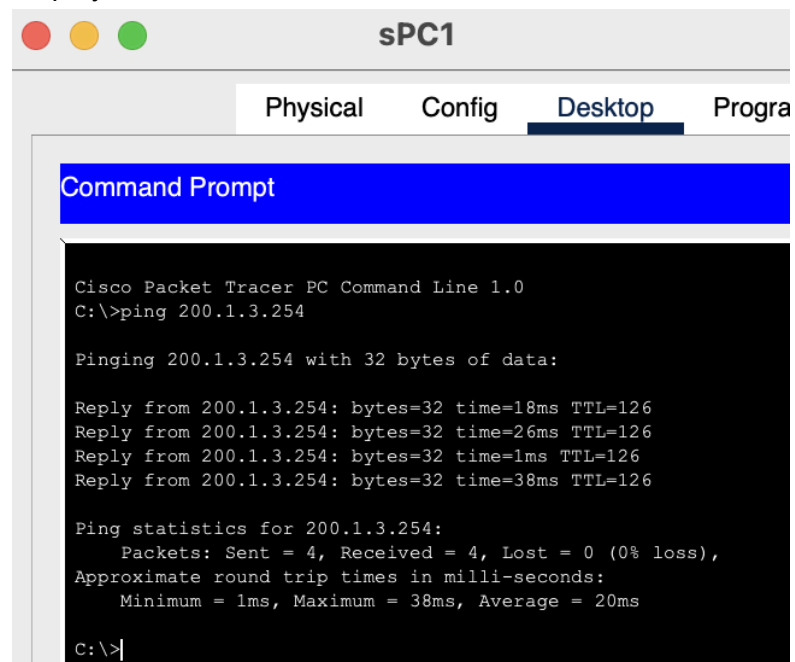
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.12.30.101

Pinging 10.12.30.101 with 32 bytes of data:

Reply from 10.12.30.101: bytes=32 time<1ms TTL=127
Reply from 10.12.30.101: bytes=32 time<1ms TTL=127
Reply from 10.12.30.101: bytes=32 time<1ms TTL=127
Reply from 10.12.30.101: bytes=32 time<1ms TTL=127

Ping statistics for 10.12.30.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

iii. Employee to Server PT outside of the network



The screenshot shows a Cisco Packet Tracer PC window titled "sPC1". The "Desktop" tab is selected. A "Command Prompt" window is open, displaying the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 200.1.3.254

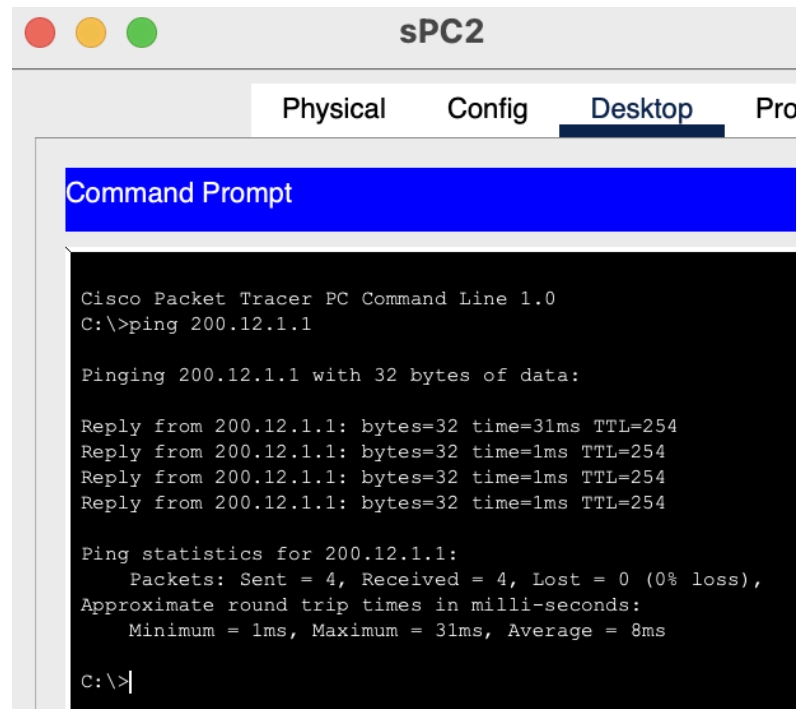
Pinging 200.1.3.254 with 32 bytes of data:

Reply from 200.1.3.254: bytes=32 time=18ms TTL=126
Reply from 200.1.3.254: bytes=32 time=26ms TTL=126
Reply from 200.1.3.254: bytes=32 time=1ms TTL=126
Reply from 200.1.3.254: bytes=32 time=38ms TTL=126

Ping statistics for 200.1.3.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 38ms, Average = 20ms

C:\>
```

iv. Employee to ISP Router



The screenshot shows a window titled 'sPC2' with tabs for 'Physical', 'Config', 'Desktop', and 'Pro'. The 'Desktop' tab is active, displaying a 'Command Prompt' window. The command prompt shows the execution of a ping command to 200.12.1.1, which is successful. The output includes the number of bytes, time, and TTL for each of the four replies, as well as summary statistics.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 200.12.1.1

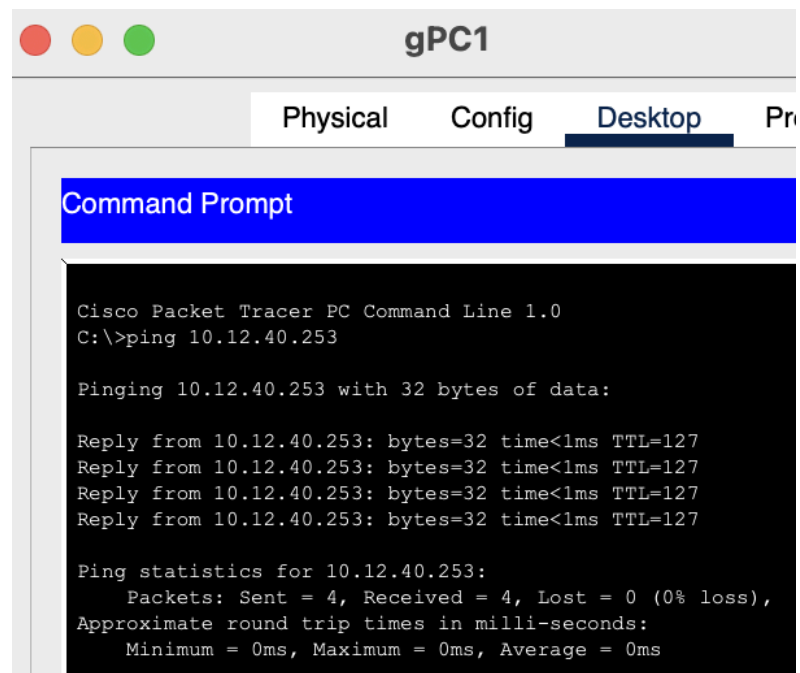
Pinging 200.12.1.1 with 32 bytes of data:

Reply from 200.12.1.1: bytes=32 time=31ms TTL=254
Reply from 200.12.1.1: bytes=32 time=1ms TTL=254
Reply from 200.12.1.1: bytes=32 time=1ms TTL=254
Reply from 200.12.1.1: bytes=32 time=1ms TTL=254

Ping statistics for 200.12.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 31ms, Average = 8ms

C:\>|
```

v. Guest to DMZ



The screenshot shows a window titled 'gPC1' with tabs for 'Physical', 'Config', 'Desktop', and 'Pro'. The 'Desktop' tab is active, displaying a 'Command Prompt' window. The command prompt shows the execution of a ping command to 10.12.40.253, which is successful. The output includes the number of bytes, time, and TTL for each of the four replies, as well as summary statistics.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.12.40.253

Pinging 10.12.40.253 with 32 bytes of data:

Reply from 10.12.40.253: bytes=32 time<1ms TTL=127
Reply from 10.12.40.253: bytes=32 time<1ms TTL=127
Reply from 10.12.40.253: bytes=32 time<1ms TTL=127
Reply from 10.12.40.253: bytes=32 time<1ms TTL=127

Ping statistics for 10.12.40.253:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

- d. What has been done: The network was segmented into 5 distinct vlans (Exec, Employee, Guest, DMZ, and NtMgmt). With the core router acting as the layer 3 device providing inter vlan communication.

2. Task2 - Services

a. DHCP (DHCP Pool & assigned IP configurations for PCs in different user groups)

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
GuestPool	10.12.30.1	10.12.40.254	10.12.30.100	255.255.255.0	50	0.0.0.0	0.0.0.0
EmployeePool	10.12.20.1	10.12.40.254	10.12.20.100	255.255.255.0	50	0.0.0.0	0.0.0.0
ExecutivePool	10.12.10.1	10.12.40.254	10.12.10.100	255.255.255.0	50	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	10.12.50.0	255.255.255.0	0	0.0.0.0	0.0.0.0

IP Configuration

☒ DHCP
 ☐ Static

IPv4 Address

Subnet Mask

Default Gateway

DNS Server

IP Configuration

☒ DHCP
 ☐ Static

IPv4 Address

Subnet Mask

Default Gateway

DNS Server

IP Configuration

☒ DHCP
 ☐ Static

IPv4 Address

Subnet Mask

Default Gateway

DNS Server

b. DNS server

No.	Name	Type	Detail
0	cs3873.net	A Record	10.12.40.253
1	cs3873.net	NS	10.12.40.254
2	cs3873.net	A Record	200.12.1.2
3	www.cs3873.net	A Record	10.12.40.253
4	www.cs3873.net	A Record	200.12.1.2

c. HTTP- Web Server and Browsing from Internal & External

HTTP

☒ On

☐ Off

HTTPS

☐ On

☒ Off

File Manager

	File Name	Edit	Delete
1	Hello.2k.html	(edit)	(delete)
2	copyrights.html	(edit)	(delete)
3	cscoptlogo177x111.jpg		(delete)
4	helloworld.html	(edit)	(delete)
5	image.html	(edit)	(delete)
6	index.html	(edit)	(delete)
7	p2.png		(delete)



- d. A single DHCP server at 10.12.50.253 serves three pools for the Executive, Employee, and Guest zones. The Core Router uses an ip helper address on each VLAN SVI to relay DHCP broadcasts to the server. DNS is hosted at 10.12.40.254 in the DMZ and resolves www.cs3873.net to 10.12.40.253 for internal hosts and 200.12.1.2 for external hosts. All three services are confirmed working via the screenshots above.

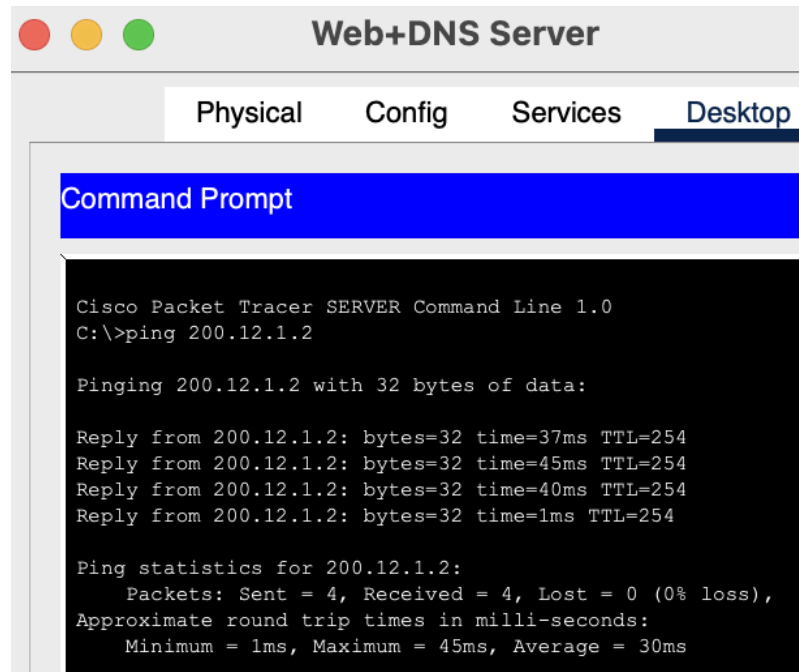
3. Task3 -NAT & PAT

```

interface Serial0/2/0
  ip address 200.12.1.2 255.255.255.252
  ip nat outside
!
a.
interface Vlan10
  mac-address 000b.bedd.9601
  ip address 10.12.10.1 255.255.255.0
  ip helper-address 10.12.50.253
  ip nat inside
!
interface Vlan20
  mac-address 000b.bedd.9602
  ip address 10.12.20.1 255.255.255.0
  ip helper-address 10.12.50.253
  ip nat inside
!
interface Vlan30
  mac-address 000b.bedd.9603
  ip address 10.12.30.1 255.255.255.0
  ip helper-address 10.12.50.253
  ip nat inside
!
interface Vlan40
  mac-address 000b.bedd.9604
  ip address 10.12.40.1 255.255.255.0
  ip nat inside
!
interface Vlan50
  mac-address 000b.bedd.9605
  ip address 10.12.50.1 255.255.255.0
  ip nat inside
!
ip nat inside source list 1 interface Serial0/2/0 overload
ip nat inside source static tcp 10.12.40.253 80 200.12.1.2 80
ip nat inside source static tcp 10.12.40.253 443 200.12.1.2 443
ip nat inside source static udp 10.12.40.254 53 200.12.1.2 53
ip nat inside source static tcp 10.12.40.254 53 200.12.1.2 53
ip default-gateway 200.12.1.1
ip classless
ip route 0.0.0.0 0.0.0.0 200.12.1.1
!
ip flow-export version 9
!
!
access-list 1 permit 10.12.0.0 0.0.255.255

```

- b. Ping request from External Web server to the CoreRouter



- c. Static NAT entries map port 80 and 443 to the internal web server at 10.12.40.253 and port 53 to the DNS server at 10.12.40.254, making only these two services directly reachable from outside the organization. Along with PAT using ACL1.

4. Task4 -Network Controller

PT-Controller0

Physical **Config** Attributes

GigabitEthernet0

Port Status

Link Speed ☐ 1000 Mbps ☒ 100 Mbps

Duplex ☐ Half Duplex ☒

MAC Address 0090.0C51.7601

IP Configuration

☐ DHCP

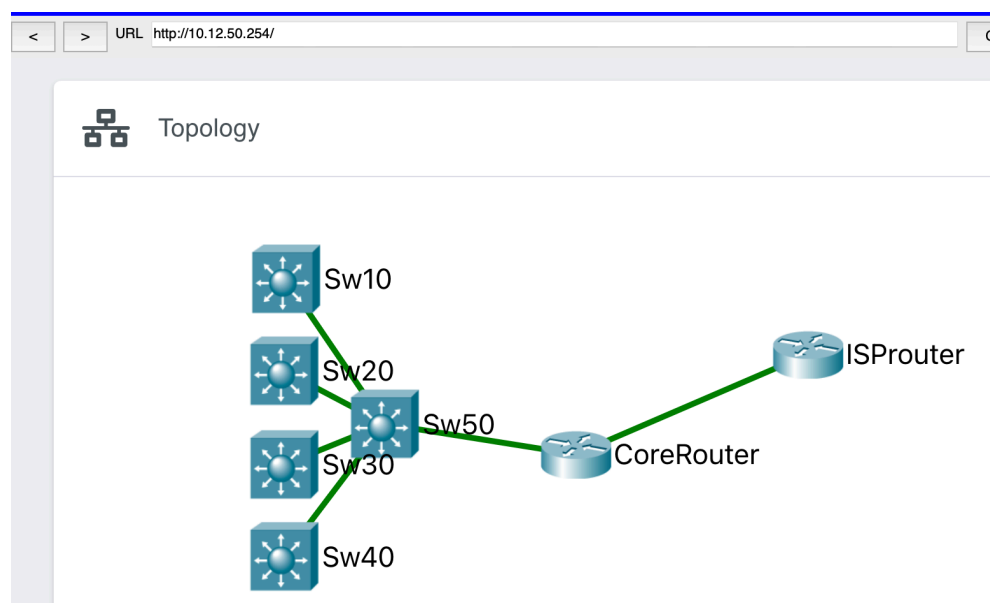
☒ Static

IPv4 Address 10.12.50.254

Subnet Mask 255.255.255.0

a.

Default Gateway 10.12.50.1





	Hostname	Type	IP	Up Time	Last U
	Sw50	MultiLayerSwitch	10.12.50.2	39 minutes, 38 seconds	2024-
	CoreRouter	Router	200.12.1.2	39 minutes, 38 seconds	2024-
	ISProuter	Router	200.12.1.1	39 minutes, 38 seconds	2024-
	Sw30	MultiLayerSwitch	10.12.30.2	39 minutes, 38 seconds	2024-
	Sw40	MultiLayerSwitch	10.12.40.2	39 minutes, 38 seconds	2024-
	Sw20	MultiLayerSwitch	10.12.20.2	39 minutes, 38 seconds	2024-
	Sw10	MultiLayerSwitch	10.12.10.2	39 minutes, 38 seconds	2024-

c.

- d. The network controller is assigned to 10.12.50.254 in VLAN50 with gateway 10.12.50.1. It has no ACL restrictions. The controller scanned the network and discovered all devices as shown in the topology screenshot. Assets are tracked and policies are implemented through the controller interface. Note that the controller is only accessible by any hosts within VLAN50 also known as the network management zone.

5. Task5 - Firewall

a. Access List on Core Router

```
CoreRouter#show access-list
Standard IP access list 1
  10 permit 10.12.0.0 0.0.255.255 (2738 match(es))
Extended IP access list v10in
  10 permit udp host 0.0.0.0 eq bootpc host 255.255.255.255 eq bootps
  20 deny ip 10.12.10.0 0.0.0.255 10.12.50.0 0.0.0.255
  30 permit ip 10.12.10.0 0.0.0.255 any
  40 deny ip any any
Extended IP access list v40in
  10 permit tcp 10.12.40.0 0.0.0.255 any established
  20 permit udp host 10.12.40.254 eq domain any
  30 permit icmp 10.12.40.0 0.0.0.255 any echo-reply
  40 permit icmp 10.12.40.0 0.0.0.255 any unreachable
  50 deny ip 10.12.40.0 0.0.0.255 any
Extended IP access list v20in
  10 permit udp host 0.0.0.0 eq bootpc host 255.255.255.255 eq bootps
  20 deny ip 10.12.20.0 0.0.0.255 10.12.50.0 0.0.0.255
  30 permit ip 10.12.20.0 0.0.0.255 any
  40 deny ip any any
Extended IP access list v30in
  10 permit udp host 0.0.0.0 eq bootpc host 255.255.255.255 eq bootps
  20 permit ip 10.12.30.0 0.0.0.255 10.12.30.0 0.0.0.255
  30 permit ip 10.12.30.0 0.0.0.255 10.12.40.0 0.0.0.255
  40 deny ip 10.12.30.0 0.0.0.255 10.12.10.0 0.0.0.255
  50 deny ip 10.12.30.0 0.0.0.255 10.12.20.0 0.0.0.255
  60 deny ip 10.12.30.0 0.0.0.255 10.12.50.0 0.0.0.255
  70 permit ip 10.12.30.0 0.0.0.255 any
```

b. BUG IN PACKET TRACER

- i. I created ACLs called v10in, v20in, v30in, and v40in. As seen by the screenshot above they were configured with the proper rules in place and were verified functional via the ping tests in the following section.
- ii. The issue is that I have a bug in packet trace where I attempt to bind the ACLs to their Vlan but the binding doesn't appear visually in the running-config file or when I show the ip interface of a vlan (demonstrated in screenshot below).
- iii. Due to not being able to visually appear in the config files, upon startup the ACLs are not applied. I understand this to be a major vulnerability within the network, so I have troubleshot by rewriting the startup-config file. Yet the bug persists with the ACLs not binding upon startup. Due to this after startup I have to run a custom script to bind the ACLs to their respective vlans.
- iv. I was preparing to replace the router with an older model such as a 2911 due to it having a fully implemented Internetwork Operating System for packet tracer, but I determined it was more important to preserve the topology set by Dr. Boppana. I also note that packet tracer is a software meant to imitate how a real network would be set up and has limitations in that representation.
- v. Below is the screenshot demonstrating the running-config that has the access-list shown, I then bind the ACLs to the proper Vlans. Note there are no errors present when binding, I then view the interface for vlan10 where it shows no inbound access list.

```

access-list 1 permit 10.12.0.0 0.0.255.255
ip access-list extended v10in
  permit udp host 0.0.0.0 eq bootpc host 255.255.255.255 eq bootps
  deny ip 10.12.10.0 0.0.0.255 10.12.50.0 0.0.0.255
  permit ip 10.12.10.0 0.0.0.255 any
  deny ip any any
ip access-list extended v40in
  permit tcp 10.12.40.0 0.0.0.255 any established
  permit udp host 10.12.40.254 eq domain any
  permit icmp 10.12.40.0 0.0.0.255 any echo-reply
  permit icmp 10.12.40.0 0.0.0.255 any unreachable
  deny ip 10.12.40.0 0.0.0.255 any
ip access-list extended v20in
  permit udp host 0.0.0.0 eq bootpc host 255.255.255.255 eq bootps
  deny ip 10.12.20.0 0.0.0.255 10.12.50.0 0.0.0.255
  permit ip 10.12.20.0 0.0.0.255 any
  deny ip any any
ip access-list extended v30in
  permit udp host 0.0.0.0 eq bootpc host 255.255.255.255 eq bootps
  permit ip 10.12.30.0 0.0.0.255 10.12.30.0 0.0.0.255
  permit ip 10.12.30.0 0.0.0.255 10.12.40.0 0.0.0.255
  deny ip 10.12.30.0 0.0.0.255 10.12.10.0 0.0.0.255
  deny ip 10.12.30.0 0.0.0.255 10.12.20.0 0.0.0.255
  deny ip 10.12.30.0 0.0.0.255 10.12.50.0 0.0.0.255
  permit ip 10.12.30.0 0.0.0.255 any
!
!
!
!
!
logging 10.12.50.253
line con 0
  password cisco123
!
line aux 0
!
line vty 0 4
  login local
  transport input ssh
!
!
!
end

```

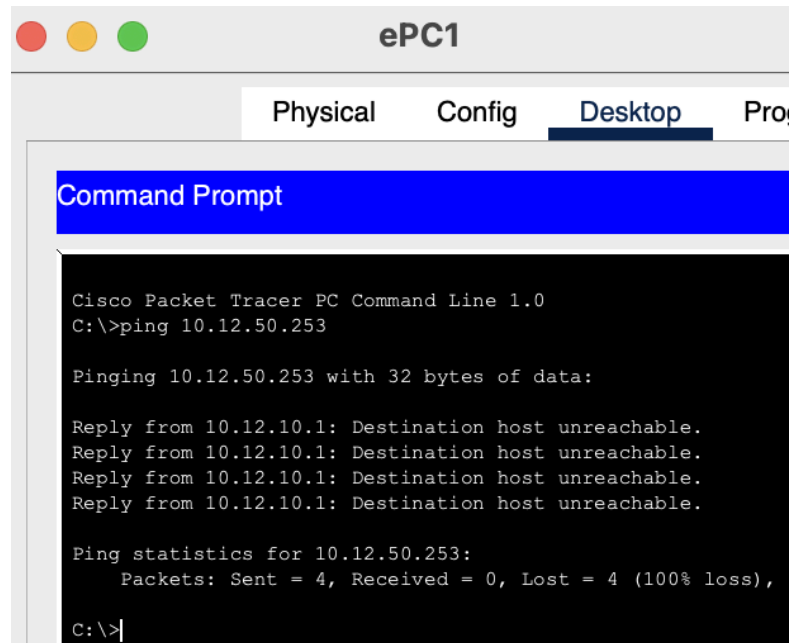
```

CoreRouter# config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
CoreRouter(config)#int vlan 10
CoreRouter(config-if)#ip access-group v10in in
CoreRouter(config-if)#int vlan 20
CoreRouter(config-if)#ip access-group v20in in
CoreRouter(config-if)#int vlan 30
CoreRouter(config-if)#ip access-group v30in in
CoreRouter(config-if)#int vlan 40
CoreRouter(config-if)#ip access-group v40in in
CoreRouter(config-if)#end
CoreRouter#show ip interface vlan 10
Vlan10 is up, line protocol is up
  Internet address is 10.12.10.1/24
  Broadcast address is 255.255.255.255
  Address determined by setup command
  MTU is 1500 bytes
  Helper address is 10.12.50.253
  Directed broadcast forwarding is disabled
  Outgoing access list is not set
  Inbound access list is not set
  Proxy ARP is enabled
  Total Proxy ARP is disabled

```

c. Failed Ping Requests

Exec to the DHCP Server in Vlan50



The screenshot shows the ePC1 interface with the Desktop tab selected. A Command Prompt window is open, displaying the output of a ping command to 10.12.50.253. The output indicates that the destination host is unreachable for all four attempts, resulting in a 100% loss of packets.

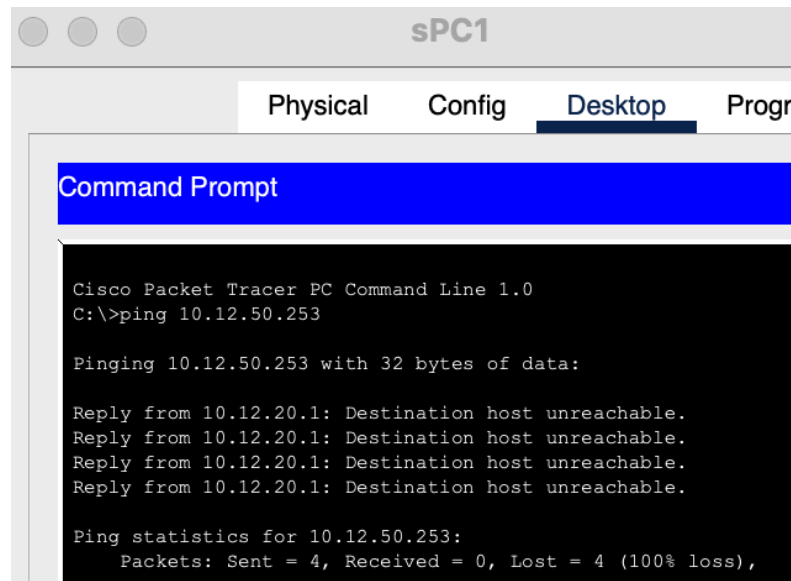
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.12.50.253

Pinging 10.12.50.253 with 32 bytes of data:

Reply from 10.12.10.1: Destination host unreachable.
Reply from 10.12.10.1: Destination host unreachable.
Reply from 10.12.10.1: Destination host unreachable.
Reply from 10.12.10.1: Destination host unreachable.

Ping statistics for 10.12.50.253:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>
```

Employee to the DHCP Server



The screenshot shows the sPC1 interface with the Desktop tab selected. A Command Prompt window is open, displaying the output of a ping command to 10.12.50.253. The output indicates that the destination host is unreachable for all four attempts, resulting in a 100% loss of packets.

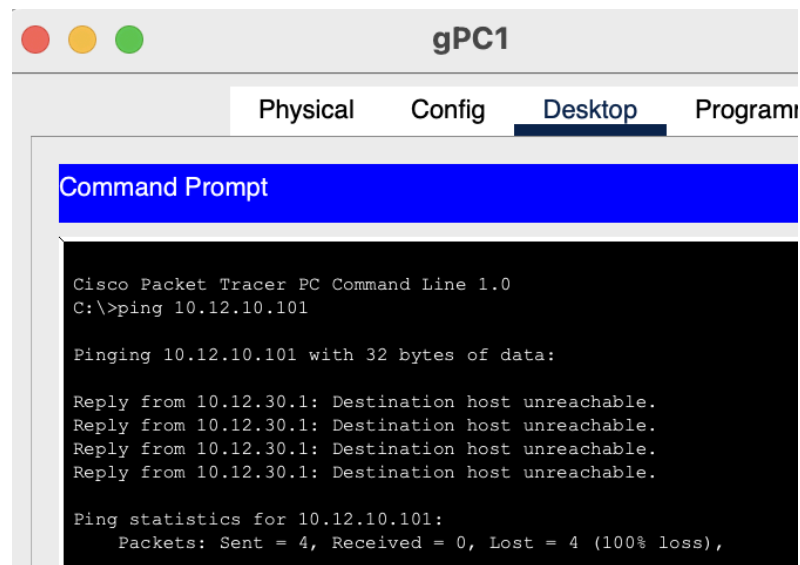
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.12.50.253

Pinging 10.12.50.253 with 32 bytes of data:

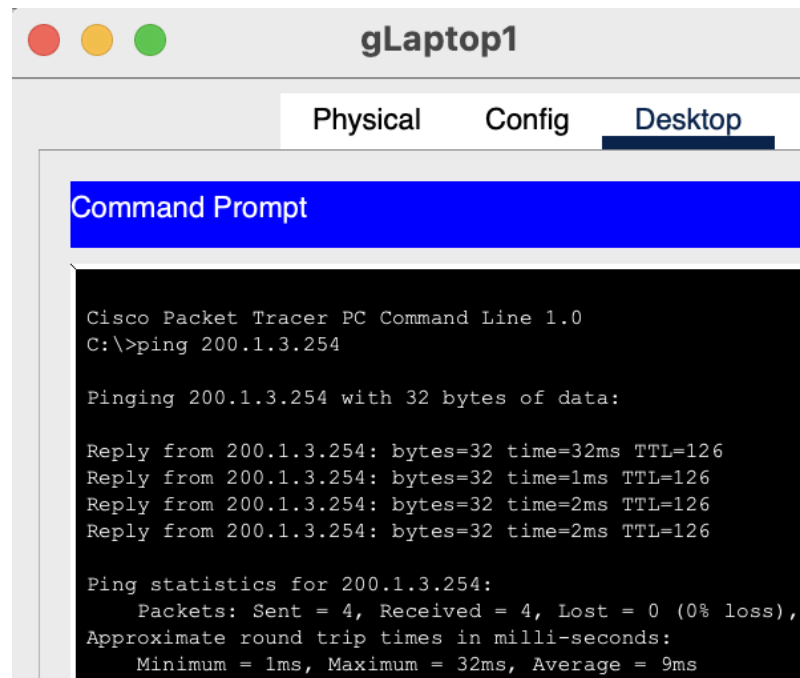
Reply from 10.12.20.1: Destination host unreachable.
Reply from 10.12.20.1: Destination host unreachable.
Reply from 10.12.20.1: Destination host unreachable.
Reply from 10.12.20.1: Destination host unreachable.

Ping statistics for 10.12.50.253:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

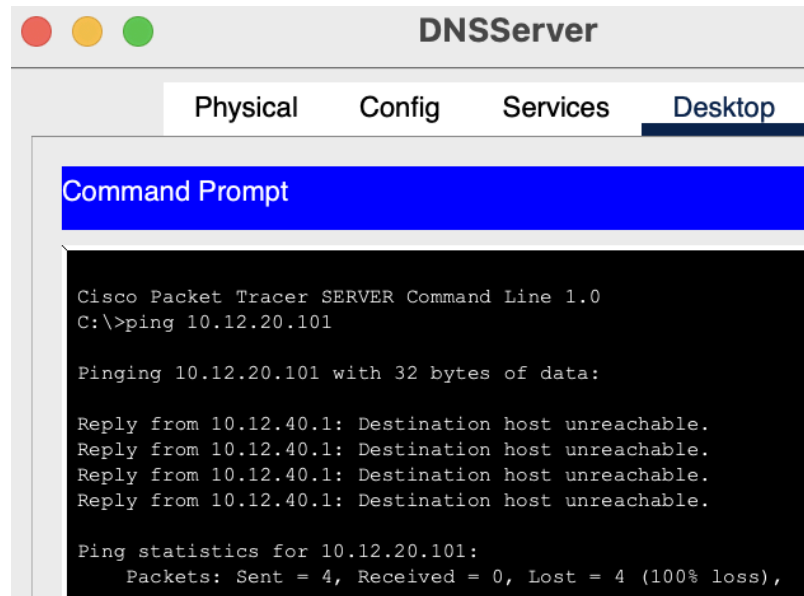
Guest to Exec User



Guest to Web+DNS Server outside the network



DNS Server to Employee



Guest to other Guest Host

